

Filter Banks That Tell the Truth: Littlewood–Paley Theory for People Who Have to Ship Code

A practitioner’s synthesis of continuous wavelet filter banks, tightness, one-integral inversion, synchrosqueezing, and scattering — assembled from the DSP.SE corpus of OverLordGoldDragon and the classical theory it rests on.

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Abstract

A filter bank is a finite or countable family of linear filters whose outputs, taken together, are supposed to represent a signal. Littlewood–Paley theory is the statement that this representation is honest if and only if a single function of frequency — the Littlewood–Paley (LP) sum — is essentially constant. That is the whole paper, said once. Everything else is the bookkeeping required to make the statement precise for continuous wavelet transforms, to invert them with one integral rather than two, to know when synchrosqueezing is licensed, and to know when a scattering network is contractive rather than merely fashionable. We collect the relevant theorems with complete proofs, translate each one into a check that can be run on a `numpy` array, and record the design rules that OverLordGoldDragon spent several years hammering into the DSP.SE record: geometric scale sampling, L^1 versus L^2 normalization, strict analyticity, zero-phase kernels, the halved Nyquist bin, and the refusal to treat “sample some scales and hope” as a wavelet transform. The aim is comprehension that survives contact with a spectrogram.

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1. What this paper is for

There is a gap, wide enough to lose a career in, between the sentence “the continuous wavelet transform is a convolution against dilated copies of a mother wavelet” and the sentence “this particular array of filters will reconstruct my signal, conserve its energy, and not lie about instantaneous frequency.” The first sentence is in every textbook. The second is what you need when you ship an analysis pipeline. The second sentence is a Littlewood–Paley statement dressed in engineering clothes.

OverLordGoldDragon’s DSP Stack Exchange corpus — roughly three hundred answers and a cluster of long-form posts on CWT inversion, filter-bank validation, synchrosqueezing, wavelet scattering, and joint time–frequency scattering [22, 23, 24] — is the most complete public attempt we know of to close that gap without pretending the analysis is optional. The present paper does three things with that corpus. First, it extracts the operational doctrine: what you actually check, in what order, and why each check is not a taste test. Second, it supplies the theorems those checks are special cases of, with proofs written so that a reader who remembers Plancherel can follow every line. Third, it places the doctrine against the classical sources — Littlewood and Paley (1931) [1], Calderón’s reproducing formula [2], Daubechies’ frames [6, 7], Mallat’s unitary condition for scattering [11], Daubechies–Lu–Wu synchrosqueezing [10] — so that a citation is never an ornament.

A word on tone. The contract here is the applied one: state the theorem, prove it, then say what a working engineer would actually do with it tomorrow morning. We will not flex about Besov indices unless a Besov index changes a design choice. We will flex about

the difference between an L^1 -normalized Morlet and an L^2 -normalized Morlet, because that difference decides whether your amplitudes mean anything.

2. The one picture you need

Fix a real line and a Fourier transform normalized as

$$\hat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i \xi t} dt, \quad f(t) = \int_{\mathbb{R}} \hat{f}(\xi) e^{2\pi i \xi t} d\xi. \quad (1)$$

Plancherel then reads $\|f\|_2 = \|\hat{f}\|_2$. A filter is a convolution against a kernel ψ , equivalently a multiplier $\hat{\psi}(\xi)$ on the Fourier side. A filter bank is a family $\{\psi_\lambda\}_{\lambda \in \Lambda}$ together with, usually, a low-pass kernel φ . The analysis operator sends f to the collection of outputs

$$(W_\lambda f)(t) = (f * \psi_\lambda)(t) = \int_{\mathbb{R}} f(u) \psi_\lambda(t - u) du. \quad (2)$$

Two questions, and only two, decide whether this object is a representation or a mood board.

Q1. Reconstruction.

Can f be recovered from the outputs, stably? In the continuous-wavelet setting the cheapest reconstruction is the one-integral formula $f \approx C^{-1} \sum_\lambda \text{Re}(W_\lambda f)$ (plus a low-pass piece). That formula is true only if the filters tile frequency.

Q2. Energy.

Is $\|f\|_2$ comparable to the ℓ^2 -sum of the output energies? If the comparison constants are equal, the bank is a tight frame and the LP sum is a constant.

Both questions collapse to the same function of frequency.

2.1 The Littlewood–Paley sum

Define the *Littlewood–Paley sum* of the bank by

$$\Phi(\xi) := |\hat{\varphi}(\xi)|^2 + \sum_{\lambda \in \Lambda} |\hat{\psi}_\lambda(\xi)|^2. \quad (3)$$

Variants exist: some authors drop the squares and work with an L^1 tiling $\sum_\lambda |\hat{\psi}_\lambda(\xi)|$; some insert a factor 2 because they keep only the positive-frequency half of an analytic bank and reconstruct the real signal from the real part of the coefficients. We will be explicit about convention every time a constant appears. The moral does not change.

If $\Phi(\xi) = A$ for almost every ξ , the bank is a tight frame with frame bound A , Plancherel holds with a single constant, and one-integral inversion is a rescaling. If $A^{-1}\Phi(\xi)$ stays between $1 - \varepsilon$ and $1 + \varepsilon$ on the frequencies you care about, you have a usable bank and you can quote the ε . If Φ has holes, those holes are frequencies the bank cannot see. If Φ has peaks, those frequencies are overweighted and your “energy in band λ ” is a lie. That is the entire practical content of Littlewood–Paley theory for filter-bank designers. Figure 1 shows both regimes.

Practitioner Note 1. OverLordGoldDragon’s validation post [24] treats the plot of Φ as the first figure you look at, not the last. Reconstruction SNR, ridge fidelity, and scattering stability are all downstream of Φ . If Φ is ugly, stop. Do not tune a synchrosqueezing hop size on a bank that does not tile; that is rearranging deck chairs on a linear scale axis.

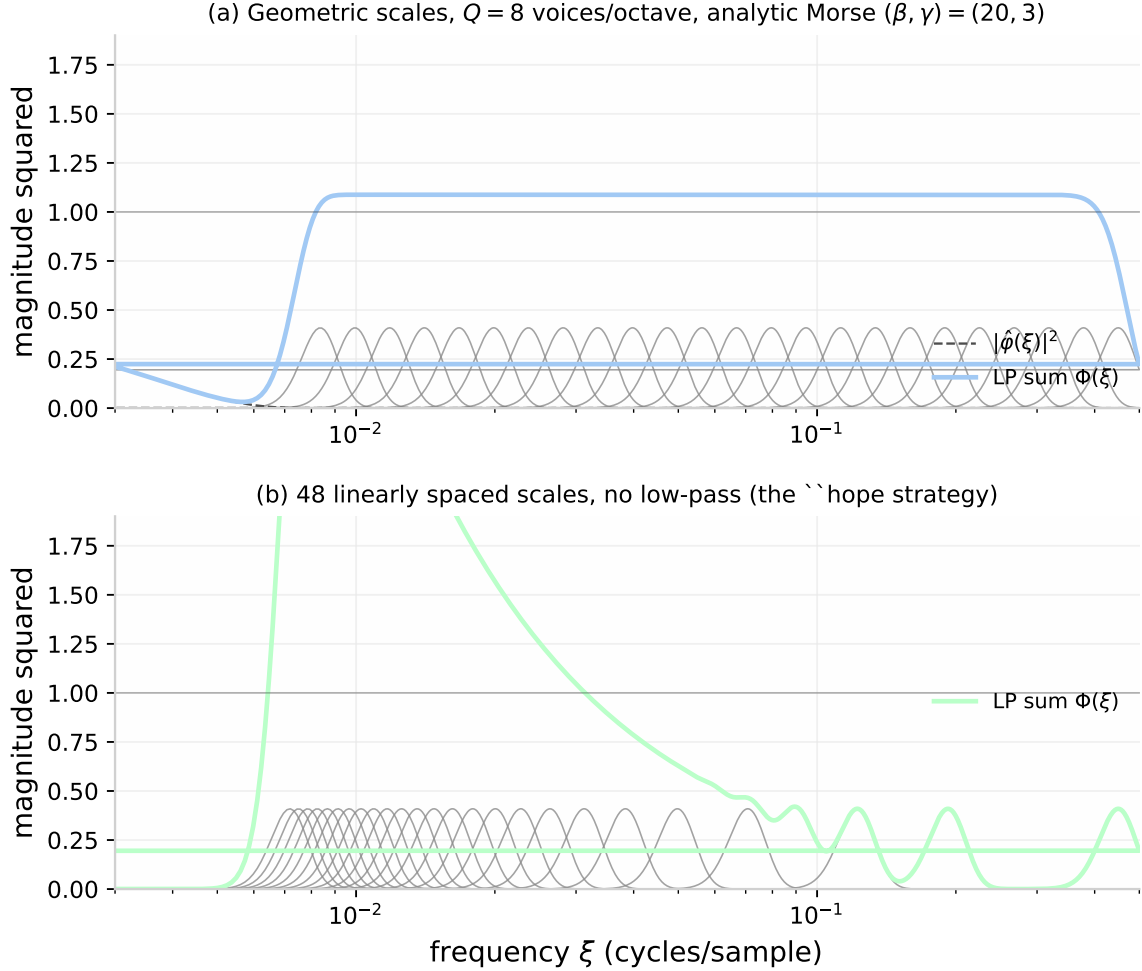


Figure 1: The entire paper in one figure. (a) A geometric bank of analytic Morse wavelets (eight voices per octave) plus a low-pass: the Littlewood–Paley sum $\Phi(\xi)$ of (3) is flat, so the bank is (numerically) tight and one-integral inversion is a rescaling. (b) The same prototype sampled at 48 linearly spaced scales with no low-pass: dense overlap at coarse scales inflates the sum, sparse sampling at fine scales leaves holes, and DC is a crater. The filters are the same; only the bookkeeping differs.

3. Littlewood–Paley theory, said properly

The original Littlewood–Paley theorem [1] (see also Edwards–Gaudry [4] and Stein [3]) is an L^p statement about dyadic blocks of a Fourier series. Let $\Delta_j f$ be the projection of f onto frequencies in the annulus $\{2^j \lesssim |\xi| \lesssim 2^{j+1}\}$. For $1 < p < \infty$ there exist constants A_p, B_p such that

$$A_p \|f\|_p \leq \left\| \left[\sum_j |\Delta_j f|^2 \right]^{1/2} \right\|_p \leq B_p \|f\|_p. \quad (4)$$

The square function $S(f) = \left(\sum_j |\Delta_j f|^2 \right)^{1/2}$ is a substitute for orthogonality when $p \neq 2$. At $p = 2$ the inequality is Plancherel and the constants collapse to 1 if the dyadic cutoff functions form a partition of unity in the L^2 sense, i.e. if $\sum_j |\hat{\varphi}(2^{-j}\xi)|^2 = 1$. That last display is the ancestor of every filter-bank LP sum in this paper.

Two remarks keep the historian and the engineer from talking past each other. First, the classical theorem is about L^p boundedness of a square function, not about perfect reconstruction.

tion of a discrete array. Second, the engineering LP sum (3) is the $p = 2$, finite-or-countable-bank specialization, which is exactly the case you can check with an FFT. We prove that case as Theorem 1 below and treat the L^p theorem as the reason the same construction remains stable when you later take moduli and cascade, as scattering does.

3.1 Calderón’s formula is a continuous LP sum

Let $\psi \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ satisfy the Calderón admissibility condition

$$0 < C_\psi := \int_0^\infty \frac{|\hat{\psi}(\xi)|^2}{\xi} d\xi < \infty. \quad (5)$$

Write $\psi_s(t) = s^{-1}\psi(t/s)$ for the L^1 -normalized dilation (we discuss L^2 -normalization in Section 5). The continuous wavelet transform is

$$(W_\psi f)(s, t) = \int_{\mathbb{R}} f(u) s^{-1} \bar{\psi}\left(\frac{u-t}{s}\right) du = (f * \tilde{\psi}_s)(t), \quad \tilde{\psi}_s(t) = s^{-1} \bar{\psi}(-t/s). \quad (6)$$

Calderón’s reproducing formula [2] then says that for f in a dense class (the Schwartz space modulo polynomials, or L^2 with a low-pass correction),

$$f = C_\psi^{-1} \int_0^\infty (W_\psi f)(s, \cdot) * \psi_s \frac{ds}{s}. \quad (7)$$

This is a two-integral inversion: you integrate in time (the inner convolution) and in scale. The one-integral inversion used in practice drops the inner convolution and sums the coefficients themselves. That drop is legal only after discretization in scale and only if the discrete LP sum is flat. Calderón gives you existence of *some* inversion. The discrete LP sum gives you the inversion you will actually type.

4. Theorems for a discrete-scale wavelet bank

We now work with a concrete object: a low-pass φ and a finite family of wavelets $\{\psi_\lambda\}_{\lambda=1}^L$, all in $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, all implemented as circular or linear convolutions of length N after sampling. On the Fourier side they are N -vectors. The continuous-frequency theorems below are the design target; the DFT versions (Section 11) are what you measure.

Theorem 1 (Plancherel identity for a filter bank). *Let $\varphi, \{\psi_\lambda\} \subset L^2(\mathbb{R})$ have Fourier transforms in L^∞ . For every $f \in L^2(\mathbb{R})$,*

$$\|\varphi * f\|_2^2 + \sum_\lambda \|\psi_\lambda * f\|_2^2 = \int_{\mathbb{R}} \Phi(\xi) |\hat{f}(\xi)|^2 d\xi, \quad (8)$$

where $\Phi(\xi) = |\hat{\varphi}(\xi)|^2 + \sum_\lambda |\hat{\psi}_\lambda(\xi)|^2$. In particular, if $A \leq \Phi(\xi) \leq B$ almost everywhere, then

$$A \|f\|_2^2 \leq \|\varphi * f\|_2^2 + \sum_\lambda \|\psi_\lambda * f\|_2^2 \leq B \|f\|_2^2. \quad (9)$$

If $A = B$, the bank is a tight frame for L^2 with frame bound A .

Proof. By Plancherel, $\|\psi_\lambda * f\|_2^2 = \|\hat{\psi}_\lambda \hat{f}\|_2^2 = \int |\hat{\psi}_\lambda(\xi)|^2 |\hat{f}(\xi)|^2 d\xi$, and likewise for φ . Sum over λ and add the low-pass term. The integrand is $\Phi(\xi) |\hat{f}(\xi)|^2$. The frame bounds are the essential infimum and supremum of Φ . Tightness is the case $\Phi \equiv A$. $\square$

Practitioner Note 2. This is the theorem you plot. Compute the DFT of each filter, square the moduli, add, and look at the result from DC to Nyquist. The ratio $\text{ess sup } \Phi / \text{ess inf } \Phi$ on the pass region is your frame condition number. Anything above about 1.2 is a bank you should redesign before you publish a figure — or at least before you show the figure to anyone who reads axes.

Theorem 2 (One-integral inversion for a zero-phase analytic bank). *Assume: (i) each ψ_λ is analytic, i.e. $\hat{\psi}_\lambda(\xi) = 0$ for $\xi < 0$; (ii) each $\hat{\psi}_\lambda$ is real-valued (zero phase) and $\hat{\varphi}$ is real-valued; (iii) the L^1 -style tiling*

$$\Psi(\xi) := \hat{\varphi}(\xi) + \sum_{\lambda} \hat{\psi}_\lambda(\xi) = 1 \quad \text{for almost every } \xi \geq 0, \quad (10)$$

and $\Psi(\xi) = 0$ for $\xi < 0$ except possibly a Hermitian completion of $\hat{\varphi}$ at the origin. Let f be real-valued and in L^2 . Then

$$f(t) = (\varphi * f)(t) + 2 \sum_{\lambda} \text{Re}(\psi_\lambda * f)(t) \quad \text{almost everywhere.} \quad (11)$$

If instead one uses the energy tiling $\Phi(\xi) = C$ on $\xi \geq 0$ with L^2 -normalized filters, the same formula holds after dividing the wavelet sum by C and treating the low-pass consistently.

Proof. Work on the Fourier side. Since f is real, $\hat{f}(-\xi) = \overline{\hat{f}(\xi)}$. Analyticity of each ψ_λ means $(\psi_\lambda * f)^\wedge(\xi) = \hat{\psi}_\lambda(\xi) \hat{f}(\xi)$ vanishes for $\xi < 0$. Taking real parts in time is the Hermitian symmetrization on the Fourier side: $2 \text{Re}(\psi_\lambda * f)^\wedge(\xi)$ equals $\hat{\psi}_\lambda(\xi) \hat{f}(\xi)$ for $\xi > 0$ and the conjugate copy for $\xi < 0$. Summing and adding the low-pass therefore multiplies $\hat{f}(\xi)$ by $\Psi(\xi)$ on the positive axis and by the Hermitian image of Ψ on the negative axis. Condition (iii) makes that multiplier identically 1. Invert the Fourier transform. $\square$

Two comments that save weeks. First, the factor 2 is not a taste: it is the price of throwing away the negative frequencies and recovering them from conjugacy. If you keep a pair of analytic and anti-analytic banks, the 2 disappears and you reconstruct complex signals. Second, zero phase is what lets you sum the time-domain outputs without a further convolution. If the filters have leftover linear phase, the sum is a delayed copy; if they have nonlinear phase, the sum is not a copy at all.

Theorem 3 (Admissibility, zero mean, and the DC hole). *If $\int \psi(t) dt = 0$, then $\hat{\psi}(0) = 0$. Consequently no finite collection of wavelets can represent the DC component of f . A low-pass φ with $\hat{\varphi}(0) \neq 0$ is mandatory for one-integral inversion on a space that contains constants. Moreover, if $\hat{\psi}$ is continuous, the vanishing at the origin forces $\Phi(0) = |\hat{\varphi}(0)|^2$; the wavelets do not help you at DC no matter how many octaves you add.*

Proof. The identity $\int \psi = \hat{\psi}(0)$ is the Fourier transform evaluated at the origin. A wavelet, by every definition that makes Calderón’s integral (5) finite near $\xi = 0$, must vanish at DC. A finite sum of continuous functions vanishing at 0 still vanishes at 0. $\square$

Practitioner Note 3. This is why “CWT of a signal, then sum over scales” without a low-pass reconstructs everything except the thing your accelerometer drift lives in. Over-LordGoldDragon is explicit: the CWT captures DC only at infinite scale, which in software is never [24].

Proposition 4 (Log-uniform tiling is constant- Q). *Suppose a generator $\hat{\psi} \geq 0$ is supported in an interval of frequencies, and the bank is the geometric sequence $\hat{\psi}_\lambda(\xi) = \hat{\psi}(\xi/a^\lambda)$ for $a > 1$. Then the LP sum $\Phi(\xi) = \sum_{\lambda} |\hat{\psi}(\xi/a^\lambda)|^2$ is a function of $\log \xi$ only, on $(0, \infty)$. In particular Φ is constant on $(0, \infty)$ if and only if the same sum, viewed as a function of $u = \log \xi$, is a periodization that adds to a constant. That happens precisely when the prototype, in log-frequency, tiles the line like an STFT window tiles linear frequency. Equivalently, the bank is constant- Q : the ratio of center frequency to bandwidth is independent of λ .*

Proof. Substitute $\xi = e^u$. Then $\hat{\psi}(\xi/a^\lambda) = \hat{\psi}(\exp(u - \lambda \log a))$, a pure translate in u . The sum of squared translates of a fixed bump is periodic in u with period $\log a$. A periodic function is constant if and only if the bump is a partition of unity (in the L^2 sense) on a single period. The support width of $\hat{\psi}(\cdot/a^\lambda)$ scales with a^λ , so bandwidth / center is invariant. $\square$

This proposition is the reason “take 64 linearly spaced scales from 1 to N ” is not a wavelet transform. Linear spacing in scale is exponential spacing in log-frequency at the fine end and a desert at the coarse end. The LP sum will show a forest of peaks at high frequency and a crater at low frequency (Figure 1b). SciPy and several PyWavelets defaults have historically done some version of this. The correction is geometric spacing, or equivalently a specified number of voices per octave, which is the constant- Q transform construction under another name [16].

5. Normalization: L^1 versus L^2 , and why it is not cosmetic

Dilation can be normalized in two ways that both appear in respectable papers and that produce numerically different arrays:

$$\psi_s^{(1)}(t) = s^{-1}\psi(t/s), \quad \psi_s^{(2)}(t) = s^{-1/2}\psi(t/s). \quad (12)$$

On the Fourier side,

$$\hat{\psi}_s^{(1)}(\xi) = \hat{\psi}(s\xi), \quad \hat{\psi}_s^{(2)}(\xi) = s^{1/2}\hat{\psi}(s\xi). \quad (13)$$

L^1 normalization preserves the L^1 norm of the kernel and therefore preserves amplitudes of sinusoids: a unit-amplitude complex exponential at the peak frequency of $\hat{\psi}$ comes out of the CWT with amplitude $|\hat{\psi}(\xi_0)|$, independent of scale. That is what you want if the number you will read off a ridge is an amplitude. L^2 normalization preserves the L^2 norm of the kernel and therefore makes Theorem 1’s frame bounds scale-invariant in the continuous-scale limit. That is what you want if the number you will read off is an energy.

Scattering networks almost always L^1 -normalize (Mallat’s convention: the wavelet is a bump of integral zero whose Fourier transform peaks near 1) and then take moduli, so that a unit sinusoid produces a unit-scale envelope [11]. Synchrosqueezing papers mix conventions; Daubechies–Lu–Wu work with a continuous family and an admissibility constant that absorbs the choice [10]. OverLordGoldDragon’s practical rule is: pick L^1 if you will invert by summing coefficients or read instantaneous amplitudes; pick L^2 if you will quote Plancherel and stop; never mix them inside one bank and then compare scales. Mixing them is how you get a figure that is simultaneously correct and meaningless, which is the worst kind of figure.

Proposition 5 (Sinusoid gain under L^1 dilation). *Let $f(t) = \cos(2\pi\xi_0 t)$. Let ψ be analytic, zero-phase, L^1 -normalized at scale s , with $\hat{\psi}(s\xi_0) = G_s$. Then*

$$(\psi_s * f)(t) = \frac{G_s}{2} e^{2\pi i \xi_0 t} + \text{the vanishing negative-frequency contribution}. \quad (14)$$

In particular, if the prototype is peak-normalized so that $\max \hat{\psi} = 1$, every scale reports the true amplitude of a tone sitting at its center frequency.

Proof. $\hat{f}$ is a pair of deltas at $\pm\xi_0$, each of mass $1/2$. Analyticity kills $-\xi_0$. The remaining delta is multiplied by $\hat{\psi}(s\xi_0) = G_s$. Inverse Fourier transform produces the displayed complex exponential, whose real part is $(G_s/2) \cdot 2 \cos = G_s \cos$ after the usual $2 \operatorname{Re}$ bookkeeping, or G_s times the analytic signal depending on how you store the coefficient. $\square$

6. Analyticity, the Hilbert tax, and the Nyquist bin

A kernel is *strictly analytic* when $\hat{\psi}(\xi) = 0$ for all $\xi < 0$. It is *pseudo-analytic* when the negative-frequency mass is small compared with the positive-frequency mass. Strict analyticity is what licenses the identification of instantaneous frequency with the phase derivative of the coefficient, which is the input to synchrosqueezing (Section 8). It is also what licenses Theorem 2’s factor-of-two reconstruction.

The cost is time support. Setting the negative frequencies to zero is a multiplication by the Heaviside function on the Fourier side, hence a convolution against a conjugate-log kernel (the Hilbert kernel, up to i) on the time side. A compactly supported real wavelet cannot be strictly analytic. Generalized Morse wavelets (Olhede–Walden [14]; Lilly–Olhede [15]) are the practical compromise: they are exactly analytic by construction, have explicit vanishing moments, and let you trade time support against frequency support with two parameters (β, γ) , where the time–bandwidth product is $P^2 = \beta\gamma$. OverLordGoldDragon’s recommendation, repeated across the Morse and validation posts, is to use Morse wavelets when the downstream consumer is an instantaneous-frequency estimator, and to accept the temporal penalty.

6.1 The halved Nyquist bin

On a length- N DFT grid the Nyquist bin $\xi = 1/2$ is its own conjugate partner. If you zero the negative frequencies and leave the Nyquist bin untouched, you have counted it once as if it were a one-sided coefficient and once as if it were a two-sided real component. The consistent analytic projector on the grid is

$$(P_+ \hat{f})[k] = \begin{cases} \hat{f}[k], & 0 < k < N/2, \\ \frac{1}{2} \hat{f}[k], & k = 0 \text{ or } k = N/2, \\ 0, & \text{otherwise,} \end{cases} \quad (15)$$

which is the DFT form of the Hilbert-transform convention that the DC and Nyquist bins of a real signal are already real and should not be doubled. Failing to halve Nyquist produces a small but visible tail in the time-domain wavelet and a corresponding slow decay that pollutes short-time localization. This is one of the “serious flaws” the validation post attributes to casual implementations [24].

6.2 Zero phase

A filter is zero-phase when $\hat{\psi}$ is real and non-negative on its pass region (or real and odd-signed in a controlled way). Equivalently the kernel is even up to a possible global sign, after a circular shift that puts $t = 0$ at the center of the array. Zero phase is what makes “sum the coefficients over λ ” coincide with “apply the LP-sum multiplier.” Any leftover linear phase is a per-band delay; any nonlinear phase is a per-band dispersion. Neither belongs in a bank you intend to invert by summation.

7. The validation protocol

The following checklist is the operational core of OverLordGoldDragon’s “How to validate a wavelet filterbank (CWT)” answer [24], restated so that each item cites a theorem above. Run it on every bank you did not generate from a published Morse or analytic Morlet constructor with geometric voices.

7.1 Analyticity

Compute the ratio of negative-frequency energy to total energy of each $\hat{\psi}_\lambda$. For a strictly analytic bank the ratio is zero to machine precision, except possibly the halved Nyquist bin. Pseudo-analytic banks should be quoted with the ratio. If you need instantaneous frequency, do not use a real wavelet and do not use a complex wavelet whose negative side is a rounding error you have not measured.

7.2 Zero mean / admissibility

Check $\hat{\psi}_\lambda(0)$ and, in time, $|\sum_n \psi_\lambda[n]|$. Both must be negligible relative to $\|\psi_\lambda\|_1$. A wavelet with leftover DC is a low-pass in denial; it will leak trend into every scale (Theorem 3).

7.3 Zero phase

Check that $\text{Im } \hat{\psi}_\lambda$ is negligible on the passband, or equivalently that the unwrapped phase is flat there. Center the kernel so that its energy centroid sits at index 0 (circular). If you built the wavelet by sampling a formula on a time grid and then FFT’d, confirm you did not introduce a linear phase by off-center sampling.

7.4 Aliasing from duration and sampling

A wavelet whose time support at coarse scale exceeds the signal length is not that wavelet; it is that wavelet wrapped around a torus, or truncated. Either way its Fourier transform is not the prototype you plotted. Two failure modes: (i) scale too large for N , so the kernel is chopped and gains DC and ripples; (ii) scale too small, so the kernel is under-sampled and aliases. The validation post treats both as first-class bugs. The fix for (i) is to cap the maximum scale at a fixed fraction of N (and let the low-pass handle the rest), or to pad. The fix for (ii) is to refuse scales whose peak frequency sits within a few bins of Nyquist without a prototype that was designed for that end of the axis.

7.5 The LP-sum plot

This is Theorem 1 made visible. Plot $\Phi(\xi)$ from 0 to Nyquist. Quote $\text{ess inf } \Phi$, $\text{ess sup } \Phi$, and the condition number on the interval between the first wavelet’s peak and the last wavelet’s peak. For an L^1 analytic bank intended for one-integral inversion, also plot the unsquared tiling $\Psi(\xi) = \hat{\varphi}(\xi) + \sum_\lambda \hat{\psi}_\lambda(\xi)$ and measure $\|\Psi - 1\|_\infty$ on $\xi \geq 0$. Reconstruction quality tracks this number (Proposition 9): if $\Psi = 1 + \varepsilon(\xi)$, the relative error on a tone at ξ is $\varepsilon(\xi)$.

7.6 Tightness versus “good enough”

A mathematically tight bank has $\Phi \equiv A$. A numerically tight bank has condition number < 1.05 over the analysis band and a designed roll-off outside it. Scattering networks often target $\Phi \leq 1$ rather than $\Phi \equiv 1$, because contractivity of the scattering propagator needs an energy bound, not an energy identity (Theorem 7). MATLAB’s `waveletScattering` documents this explicitly: the plotted LP sums “do not exceed 1” [26]. That is a feature of the contractive design, not a failure to tile. Know which theorem you are serving.

7.7 Reconstruction test

Take a unit impulse, a unit tone at a mid-band frequency, a unit tone near the lowest wavelet, a unit tone near Nyquist, and a chirp that sweeps the band. Invert by Theorem 2. Report relative L^2 error. If the LP sum predicted a 3% ripple and you measure a 30% error, you

have a phase, padding, or normalization bug, not an LP-sum bug. The two tests diagnose different sins.

8. Synchrosqueezing, licensed by a good bank

Synchrosqueezing (Daubechies–Lu–Wu [10]; Thakur–Wu and follow-ups [17]) is not a different transform. It is a remapping of an existing CWT. Given $W(s, t) = (W_\psi f)(s, t)$, define the phase transform wherever $W \neq 0$ by

$$\omega(s, t) = \frac{1}{2\pi} \partial_t \arg W(s, t) = \operatorname{Im} \left(\frac{\partial_t W(s, t)}{2\pi W(s, t)} \right). \quad (16)$$

For a pure tone $f(t) = Ae^{2\pi i \xi_0 t}$ and an analytic wavelet, $W(s, t) = A \hat{\psi}(s \xi_0) e^{2\pi i \xi_0 t}$, so $\omega(s, t) = \xi_0$ wherever the coefficient is nonzero. The energy that the CWT had spread along the scale axis at each t is then reassigned to the frequency coordinate $\omega(s, t)$:

$$T(\eta, t) = \int_{s: |\omega(s, t) - \eta| < \varepsilon} W(s, t) s^{-3/2} ds \quad (17)$$

(the exact Jacobian in s depends on the L^1 versus L^2 convention; the display is schematic). The point, for this paper, is the hypothesis the calculation used: analyticity, so that a tone produces a coefficient whose phase is exactly $2\pi \xi_0 t$, not a mixture of ξ_0 and $-\xi_0$. A real wavelet on a real tone produces a standing-wave coefficient whose phase derivative is not the frequency. Synchrosqueezing a non-analytic bank is a well-defined operation that computes a well-defined mess.

Proposition 6 (Phase transform of an isolated tone). *Let ψ be strictly analytic, $f(t) = Ae^{2\pi i \xi_0 t}$ with $A \neq 0$, and let s be such that $\hat{\psi}(s \xi_0) \neq 0$. Then $W(s, t)$ never vanishes and $\omega(s, t) = \xi_0$ for all t .*

Proof. $W(s, t) = A \hat{\psi}(s \xi_0) e^{2\pi i \xi_0 t}$ by the convolution theorem and analyticity (the negative-frequency delta is killed). The prefactor is a nonzero constant. Differentiating the argument in t yields $2\pi \xi_0$. $\square$

OverLordGoldDragon’s long synchrosqueezing explainer [22] emphasizes the same pipeline the theorem suggests: build an analytic filter bank whose LP sum is flat, compute W , compute ω by a stable discrete derivative (often a forward difference of the unwrapped phase, or a quotient of differentiated and undifferentiated CWT, which is more stable near zeros), reassign, and invert by summing the remapped coefficients along frequency, which is legal because reassignment is designed to preserve the one-integral inversion of Theorem 2 for isolated AM–FM components. The inversion is not a general-purpose inverse; it is an inverse under the sparse AM–FM model the method assumes. If your signal is a broadband transient, synchrosqueezing will cheerfully reassign the transient’s energy to a cartoon frequency and you will deserve the cartoon.

9. Scattering: the LP condition as a contractivity hypothesis

Mallat’s group-invariant scattering [11] iterates a wavelet modulus and a low-pass. Write $U[\lambda]f = |f * \psi_\lambda|$ and $A_J f = f * \varphi_J$. A path $p = (\lambda_1, \dots, \lambda_m)$ produces

$$S[p]f = A_J U[\lambda_m] \cdots U[\lambda_1]f. \quad (18)$$

The windowed scattering concatenates the zeroth-order term $A_J f$ with all first-order terms $A_J U[\lambda]f$ and a truncated list of higher-order paths. Two properties make the object more

than a fixed CNN: it is nonexpansive on L^2 , and it is Lipschitz with respect to small C^2 diffeomorphisms (time-warps), with a Lipschitz constant that degrades only as a moment of the warp’s gradient. Both properties use the filter bank being a semi-discrete Parseval frame — that is, an LP condition.

Theorem 7 (Mallat’s unitary (Littlewood–Paley) condition [11]). *Let $\{\psi_\lambda\}$ be a wavelet family indexed by scale and, in several dimensions, rotation, and let φ_J be the low-pass at scale 2^J . The operator $W_J f = \{A_J f, (f * \psi_\lambda)_{\lambda \in \Lambda_J}\}$ is an isometry on $L^2(\mathbb{R}^d; \mathbb{C})$ if and only if*

$$|\hat{\varphi}_J(\xi)|^2 + \sum_{\lambda \in \Lambda_J} |\hat{\psi}_\lambda(\xi)|^2 = 1 \quad a.e. \quad (19)$$

On real-valued L^2 the same identity holds with a factor 1/2 on the wavelet side if the wavelets come in conjugate pairs that split positive and negative frequencies (Mallat’s β -convention: $\beta = 1$ complex, $\beta = 1/2$ real). This is Proposition 2.1 of [11].

Proof. W_J is a multiplier operator on the Fourier side. Its squared operator norm at frequency ξ is exactly the displayed sum. W_J is an isometry iff that multiplier has modulus 1 almost everywhere, iff the sum equals 1. The real-valued case follows by restricting to Hermitian $\hat{f}$ and counting each conjugate pair once. $\square$

Nonexpansiveness of scattering is then a one-line corollary. The modulus is nonexpansive on $\mathbb{C}$, hence on L^2 . A composition of nonexpansive maps with an isometry (or a contraction, if one designs $\Phi \leq 1$ rather than $\Phi = 1$) is nonexpansive. Energy is pushed to coarser scales by the modulus — this is the content of the “admissible wavelet” condition in Mallat’s Theorem 2.6, which asks that a low-pass envelope dominate the iterated wavelet energy so that the tail of the path tree decays. Numerically the decay is observed to be exponential in audio and images, which is why classification pipelines stop at order $m = 2$ or $m = 3$ [12].

Practitioner Note 4. MATLAB’s `littlewoodPaleySum` method on a `waveletScattering` or `timeFrequencyScattering` object is exactly Theorem 7 made into a method [26]. The documentation’s remark that the theoretical best case is a straight line at height 2 from 0 to Nyquist (JTFS, analytic pairs) or height 1 (contractive scattering) is the same remark as (3) of this paper. Energy-corrected filters are a numerical projection onto the constraint $\Phi \approx$ constant.

9.1 Joint time–frequency scattering

Joint time–frequency scattering (Andén–Lostanlen–Mallat [13]) applies a second wavelet transform not only along time but along the log-frequency axis of the first-order scalogram. The LP condition is imposed twice: once on the time filter banks (first and second order) and once on the frequency filter banks (spin-up and spin-down). The spin $\pm$ structure is the analytic pair along log-frequency; it is what lets JTFS distinguish upward from downward chirps, which ordinary time-scattering cannot. The validation discipline does not change. If the frequency-wavelet LP sum is a sawtooth, the “joint” part of JTFS is a joint mess.

10. Design recipes that survive the theorems

10.1 A CWT bank you can invert

1. Choose an analytic prototype with known vanishing moments. Generalized Morse with the standard choice $\gamma = 3$ and time–bandwidth product $P^2 = \beta\gamma = 60$ — i.e. $(\beta, \gamma) = (20, 3)$, the MATLAB default — is a workhorse [14, 15, 26]; analytic Morlet with a large enough carrier ($\omega_0 \geq 6$) is acceptable if you zero the residual negative frequencies and renormalize.

2. Space scales geometrically: $s_\lambda = s_{\min} \cdot 2^{\lambda/Q}$, with Q the voices per octave, typically 8–16 for analysis, 4–8 for scattering.
3. L^1 -normalize and peak-normalize the prototype so $\max \hat{\psi} = 1$.
4. Build a low-pass φ whose $\hat{\varphi}$ fills the hole that the coarsest wavelet leaves at DC, with $\hat{\varphi}$ real, decreasing, and $\hat{\varphi}(0) = 1$.
5. Optionally apply an energy correction: multiply each $\hat{\psi}_\lambda$ by a slowly varying function of λ so that Φ flattens. This is a diagonal preconditioner for Theorem 1. It slightly perturbs Q ; measure Q after, not before.
6. Run Section 7. Publish the LP-sum plot next to the scalogram, the way you would publish a calibration curve next to a measurement.

10.2 A scattering bank that is actually contractive

Target $\Phi(\xi) \leq 1$, not $\Phi(\xi) = 1$. The inequality is what gives $\|Sf - Sg\|_2 \leq \|f - g\|_2$. If Φ overshoots, a two-tone input at the overshoot frequency can increase energy under the modulus and the network is no longer a contraction. Mallat’s theory [11], and the MATLAB default `EnergyCorrectFilters = true` [26], are both serving this inequality. Second-order filters are allowed to be broader (smaller Q) because the input to the second order is already an envelope; spending voices there has diminishing returns and hurts the LP sum.

10.3 What not to do

Do not linearly space scales and call the result a CWT. Do not use a real wavelet to estimate instantaneous frequency. Do not invert by summing coefficients from a bank whose filters were L^2 -normalized and then treat the output as a voltage. Do not trust a library default: SciPy’s `cwt` historically sampled a width parameter in a way that does not implement a constant- Q LP tiling; PyWavelets’ `cwt` is closer but still requires you to pass scales you designed; `ssqueezepy` [25] exists precisely because the defaults elsewhere were not designed. Do not skip the impulse-reconstruction test. Do not synchrosqueeze a bank you have not validated. Do not report scattering coefficients from a network whose `littlewoodPaleySum` you have never plotted.

11. Worked identities the implementer uses

11.1 DFT form of Theorem 1

On $\mathbb{C}^N$ with the unitary DFT,

$$\sum_n |(\varphi * f)[n]|^2 + \sum_\lambda \sum_n |(\psi_\lambda * f)[n]|^2 = \sum_k \Phi[k] |\hat{f}[k]|^2, \quad (20)$$

where $\Phi[k] = |\hat{\varphi}[k]|^2 + \sum_\lambda |\hat{\psi}_\lambda[k]|^2$ and convolution is circular. If you use linear convolution via padding, Plancherel holds on the padded grid and you must restrict Φ to the unpadded frequencies when you quote bounds. Mixing the two is a standard way to get a “mystery” 3 dB.

11.2 The one-integral formula on the grid

For a real input $x[n]$, analytic zero-phase bank, and unsquared tiling $\Psi[k] \approx 1$ on $k = 0, \dots, N/2$,

$$x[n] \approx (\varphi * x)[n] + 2 \sum_\lambda \operatorname{Re}(\psi_\lambda * x)[n]. \quad (21)$$

If the bank is stored as rows of a matrix Ψ_{time} of shape (L, N) , this is $x \approx \text{lowpass} + 2 \operatorname{Re}(\Psi_{\text{time}} x)$ summed over rows, implemented as an FFT multiply by the precomputed $\hat{\psi}_\lambda$. That is the entire inverse CWT in the one-integral convention. Two-integral inversion convolves again with the synthesis wavelets and is the discrete Calderón formula; it is more forgiving of a non-flat Ψ at the price of a second filter-bank pass.

11.3 Scale-to-frequency

For an L^1 -normalized wavelet with peak frequency ξ_ψ of the prototype (in cycles per sample at scale 1), the peak at scale s sits at ξ_ψ/s . Convert to hertz by multiplying by the sampling rate. Do not use the band-limited center of mass and the peak interchangeably without checking; Morse wavelets with large β have mass noticeably to the left of the peak. Quote which one you used when you label a scalogram axis.

12. Frames, and why tightness is the friend of inversion

We record the abstract frame lemma that Theorem 1 specializes, because synchrosqueezing and scattering both quietly rely on it.

Lemma 8 (Dual frames and stable inversion). *Let $\{\psi_\lambda\} \subset H$ be a frame for a Hilbert space H with bounds $0 < A \leq B < \infty$, i.e. $A\|f\|^2 \leq \sum_\lambda |\langle f, \psi_\lambda \rangle|^2 \leq B\|f\|^2$. Then the frame operator $Sf = \sum_\lambda \langle f, \psi_\lambda \rangle \psi_\lambda$ is bounded, positive, and invertible, with $AI \leq S \leq BI$, and*

$$f = \sum_\lambda \langle f, \psi_\lambda \rangle S^{-1} \psi_\lambda = \sum_\lambda \langle f, S^{-1} \psi_\lambda \rangle \psi_\lambda. \quad (22)$$

If $A = B$ the frame is tight, $S = AI$, and the dual frame is $\{A^{-1}\psi_\lambda\}$. Analysis coefficients invert by the same family that produced them.

Proof. The quadratic form $\langle Sf, f \rangle = \sum_\lambda |\langle f, \psi_\lambda \rangle|^2$ lies between $A\|f\|^2$ and $B\|f\|^2$ by the frame inequality, so S is positive and bounded and $\operatorname{spec}(S) \subset [A, B]$. Hence S is invertible and $\operatorname{spec}(S^{-1}) \subset [B^{-1}, A^{-1}]$. The displayed reconstructions are $S^{-1}Sf = f$ and $SS^{-1}f = f$. Tightness makes S a scalar. $\square$

In the Fourier-multiplier setting of this paper, S is multiplication by Φ , and S^{-1} is multiplication by $1/\Phi$. One-integral inversion with a non-flat Φ is possible: divide each $\hat{\psi}_\lambda$ by Φ on the Fourier side and you have implemented S^{-1} . That is the theoretically correct “energy correction.” Doing the correction once, offline, on the filters, is cheaper than dividing by Φ per signal, and is what the scattering toolkits do. If Φ has zeros you cannot invert; those zeros are the holes in the LP sum, and no dual-frame algebra will invent information the analysis never measured.

12.1 A quantitative reconstruction bound

Proposition 9 (Reconstruction error tracks LP ripple). *In the setting of Theorem 2, suppose the unsquared tiling satisfies $|\Psi(\xi) - 1| \leq \varepsilon$ on the support of $\hat{f}$, with Ψ real. Then the one-integral reconstruction f_{rec} obeys*

$$\|f - f_{\text{rec}}\|_2 \leq \varepsilon \|f\|_2. \quad (23)$$

Proof. On the Fourier side, $\hat{f}_{\text{rec}} - \hat{f} = (\Psi - 1)\hat{f}$ after the analytic bookkeeping of Theorem 2. Take L^2 norms and use $|\Psi - 1| \leq \varepsilon$. $\square$

This is the reason the validation post says reconstruction quality can be predicted from the LP sum [24]. It is not a slogan. It is Proposition 9.

13. Historical notes, kept short

Littlewood and Paley [1] introduced the square-function comparison for Fourier series; the point was to move L^p questions about multipliers onto an orthogonal-looking object. Calderón [2] gave the continuous reproducing formula that wavelet theorists still cite as admissibility. Grossmann–Morlet [5] and Daubechies [6, 7] turned the formula into frames and orthonormal bases. Mallat [8] connected the bases to iterated filter banks and multiresolution. The engineering LP sum as a plot you stare at is older than scattering — every QMF designer computed a power-complementary identity [18, 19] — but Mallat [11] made the identity the hypothesis of a stability theorem for a nonlinear cascade, which is why MATLAB now ships `littlewoodPaleySum` as a method on a scattering object rather than as a remark in a wavelet chapter [26].

Daubechies–Lu–Wu [10] put synchrosqueezing on a theorem-level footing and made analyticity non-negotiable. Lilly–Olhede and Olhede–Walden [15, 14] gave the Morse family that lets an implementer obey analyticity without pretending a truncated Gabor function is a wavelet. OverLordGoldDragon’s contribution, which is the reason this paper exists, was to refuse the idea that these theorems live in a different room from the `numpy` array. The validation report, the insistence on geometric voices, the halved Nyquist bin, the public comparison of SciPy against a bank that actually tiles — that is the missing chapter in most CWT tutorials [22, 23, 24, 25].

14. What we did not cover, on purpose

Orthonormal discrete wavelet bases and the two-channel perfect-reconstruction identity $H_0(z)G_0(z) + H_1(z)G_1(z) = cz^{-m}$ are a different and fully solved subject [19, 18]. They satisfy an LP identity by construction, at the price of critical sampling and shift variance. This paper is about redundant banks, because CWT, synchrosqueezing, and scattering are redundant on purpose. Dual-tree complex wavelets, curvelets [21], shearlets, and wave-packet scattering [12] all have their own LP sums; the algebra is Theorem 1 with a different index set λ .

Higher-dimensional Littlewood–Paley theory, UMD spaces, and multiplier theorems on L^p [3, 4] are the reason analysts still write papers with these names on them. They do not change the plot you owe your reader on page two of an experimental supplement.

15. A closing inventory

If you remember four displays, you can reconstruct the paper.

The LP sum. $\Phi(\xi) = |\hat{\varphi}(\xi)|^2 + \sum_{\lambda} |\hat{\psi}_{\lambda}(\xi)|^2$. Flatness is tightness. Holes are blindness. Peaks are lies about energy.

One-integral inversion. For a real signal and an analytic zero-phase bank with unsquared tiling $\Psi \equiv 1$ on $\xi \geq 0$: $f = \varphi * f + 2 \sum_{\lambda} \text{Re}(\psi_{\lambda} * f)$.

Geometric voices. Dilation is translation in log-frequency. Linear scale grids do not tile. Constant- Q is not a musical preference; it is Proposition 4.

License for the nonlinear steps. Synchrosqueezing needs analyticity so that $\omega(s, t)$ equals the tone frequency. Scattering needs $\Phi \leq 1$ so that the cascade is a contraction. Both inherit their honesty from the linear bank you built first.

The rest is hygiene: zero mean, zero phase, halved Nyquist, duration versus scale, L^1 versus L^2 , and a reconstruction test that you do not skip because the plot “looked like a wavelet.” Littlewood–Paley theory, in the only sense a practitioner needs, is the decision to measure that hygiene against a theorem rather than against a vibe.

Colophon. This paper is a reading of a public corpus, not a co-authorship. Errors of translation from DSP.SE idiom into theorem-and-proof idiom are the present author’s. The doctrine that a filter bank is only as good as its Littlewood–Paley sum is OverLordGold-Dragon’s, and it is correct.

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